In [8]:
```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
```

In [9]:
```
df = pd.read_csv(r"C:\Users\SINAN\IRIS.csv")
df
```

Out[9]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Iris-virginica
146	6.3	2.5	5.0	1.9	Iris-virginica
147	6.5	3.0	5.2	2.0	Iris-virginica
148	6.2	3.4	5.4	2.3	Iris-virginica
149	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 5 columns

In [10]:
```
X = df[['sepal_length', 'sepal_width', 'petal_length', 'petal_width']]
Y = df['species']
```

In [11]:
```
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state=42)
```

In [12]:
```
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

In [13]:
```
k = 5
knn = KNeighborsClassifier(n_neighbors=k)
knn.fit(X_train, Y_train)
```

Out[13]:

KNeighborsClassifier

▶ Parameters

In [15]:
```
Y_pred = knn.predict(X_test)
```

In [16]:
```
print("Accuracy:", accuracy_score(Y_test, Y_pred))
print("\nConfusion Matrix:\n", confusion_matrix(Y_test, Y_pred))
```

```
Accuracy: 1.0

Confusion Matrix:
 [[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```